SCREEN-TIME BANK · ONE HTML FILE · OFFLINE PWA · APK · VANILLA JS · NO DEPS · PYTEST + PLAYWRIGHT

The Marbles

Kids earn real marbles for jobs around the house. This app is the piggy bank and the timer: marbles buy game time in 10-minute steps, spare ones are saved for bigger rewards, and grown-ups manage it all behind a PIN.

1	0	5 min	30+
file: the whole app is `index.html`	runtime libraries, build steps or network calls	pause budget per session, then it restarts itself	behaviour tests driving the real page in Chromium

THE STACK

Plain web platform, on purpose

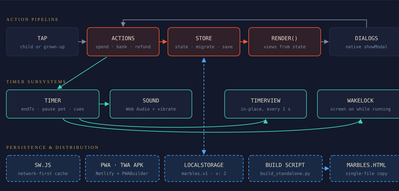
Everything a family phone already has: HTML, CSS, JavaScript and a few browser APIs. The chart and the marbles are hand-drawn SVG; the sounds are synthesised.

APP	STATE	TIMER	SOUND
index.html HTML, CSS and JS in one file, split internally into single-job modules.	**localStorage** One JSON object under `marbles.v1`, schema `v: 2`, with migration.	**Wall clock** `endTs`-based, so it survives reloads and screen locks.	**Web Audio** Synthesised tones plus vibrate; no audio files to load.
SCREEN	DIALOGS	OFFLINE	ANDROID
Wake Lock Keeps the screen on while a timer runs so the alarm can ring.	**<dialog>** `showModal()` for focus trap, Esc and back-to-close.	**sw.js** Network-first pages with a cached fallback; updates land on the next launch.	**PWABuilder TWA** Signed `.apk` that loads the hosted site; no rebuild for app changes.

ARCHITECTURE

One file, clear lanes

A tap calls an action, the action changes the store, and `render()` rebuilds the page from state. The timer is the one exception: it updates only the clock, ring and labels in place, once a second.



Ember arrows are the main loop: action → store → full render. The grey loop back from dialogs feeds grown-up choices into actions. Teal arrows are the timer lane: spending starts the timer, which drives sound, the in-place clock and the wake lock without redrawing the page. The dashed sky line is persistence: the store loads from and saves to `localStorage`; the other boxes show how the same file ships, including the generated single-file copy.

HOW IT WAS BUILT

Six phases

From the data model out to the phone's home screen. Open a phase to see its steps.

01 Store & data model — STORE

One JSON object, safe to evolve

- 1.1 **Schema v2** — Marbles, activities, goals, a 300-entry history and the running session in one object.
- 1.2 **v1 → v2 migration** — Rates snapped to 10, history repaired, old session shape converted, `setupDone` inferred.
- 1.3 **Persistent storage** — The app asks the browser not to evict its data when the phone is low on space.
- 1.4 **Backup codes** — Base64 JSON without the PIN; restore shows now vs backup and keeps the current PIN.

02 Spending rules — ACTIONS

Fair by construction

- 2.1 **Cost from the rate** — `ceil(minutes + minutes-per-marble)`, always computed inside `spendMarble`, never passed in.
- 2.2 **No borrowing** — Overspending and a second concurrent timer are refused.
- 2.3 **Same-price offer** — When the chosen time is under what the marbles pay for, a button offers the full paid-for minutes.
- 2.4 **Refund on Done** — Every whole marble of unused time comes back; time played past zero is recorded.

03 Timer engine — TIMER

Reliable on a phone that sleeps

- 3.1 **Wall-clock timing** — An end timestamp, not a counter, so reloads and screen locks don't lose time.
- 3.2 **Pause pot** — Five minutes of pause per session; when it runs out the clock resumes by itself.
- 3.3 **Cues, once each** — A note at 1 min, two at 10 s, chime and buzz at zero; after the background only the most urgent plays.
- 3.4 **In-place ticks** — `TimerView.update()` touches only clock, ring and labels, so nothing else redraws.
- 3.5 **Wake lock** — Screen stays on while running, visible and not paused; re-requested on return.

04 Bank & goals — BANK

Saving toward something bigger

- 4.1 **Add and take back** — Marbles move into a goal one at a time and can be withdrawn.
- 4.2 **Claim** — A Claim button appears when a goal is full.
- 4.3 **Nothing silently lost** — Deleting a goal or cutting its price refunds the banked marbles.

05 Grown-ups, PIN & usage — PIN

Locked, calm, descriptive

- 5.1 **First-run PIN** — Setup can't be dismissed and refuses `1234`; old installs must change it.
- 5.2 **Lockout backoff** — Five wrong tries lock the keypad for 1, 2, 4… up to 15 minutes; the lock survives reloads.
- 5.3 **Grace window** — Three minutes unlocked after a correct PIN; leaving the app locks again.
- 5.4 **Usage chart** — 14 days of minutes actually played plus marbles in and out, hand-drawn SVG with finger-sized tap columns.
- 5.5 **Accessibility** — Zoom allowed, real buttons with `aria-pressed`, no clipping at 320 px.

06 Ship it — DEPLOY

Browser, home screen, APK

- 6.1 **PWA** — Static manifest, icons and a network-first service worker; deploy with Netlify Drop.
- 6.2 **Android APK** — PWABuilder wraps the live site as a Trusted Web Activity.
- 6.3 **Keystore hygiene** — `.gitignore` blocks `*.keystore` and `*.apk`; future builds reuse the same key.
- 6.4 **Standalone file** — `build_standalone.py` inlines the icon and drops the service worker to make `marbles.html`.

TEST STRATEGY

What keeps it fair

Every check drives the real `index.html` in Chromium: `pytest tests -q`.

RULES — pytest · playwright

- **Spending**: cost maths, 10-minute snapping, same-price offer
- **Refusals**: overspending and a second timer
- **Bank**: add, take back, claim, refunds on delete or price cut

TIMER — clock control in the page

- **Pause budget** auto-resumes; manual resume spends it
- **Cues** play in order, and only one after the background
- **No full redraw** on a tick; survives a reload
- **Done** confirms, refunds and records minutes played

DATA & SECURITY — state seeded via localStorage

- **PIN**: setup, forced change, lockout across reload, grace window
- **Backup** excludes the PIN; restore keeps it; old codes work
- **Migration** from v1 data to v2
- **Standalone**: a test fails if `marbles.html` is stale

MANUAL — on the family phone

- Install from Chrome and from the APK
- Timer to zero with the screen left on: sound and buzz
- Airplane mode launch uses the cached copy
- Redeploy, relaunch online, confirm the update

IN THE REPO

Where things live

`index.html`	The entire app, hosted PWA version. Edit this one.
`marbles.html`	Generated single-file copy that works from disk.
`build_standalone.py`	Builds `marbles.html` from `index.html`.
`sw.js` · `manifest.webmanifest`	Offline cache and install metadata.
`tests/test_marbles.py`	Playwright behaviour tests.
`DOCS.md`	Full technical reference.